Flexible control of intrusive thoughts depends on multiple inhibitory mechanisms

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1 INTRODUCTION

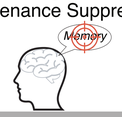
Background

- Inhibitory control can be exerted over memory representations at different stages of processing, reducing long- or short-term access[1,2].

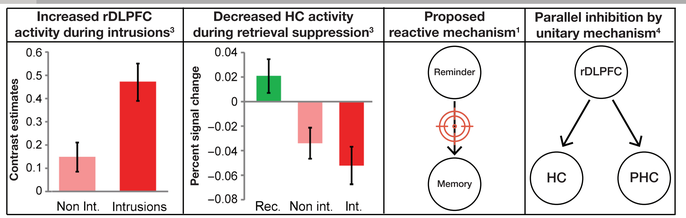
Long-Term Memory ("Retrieval Suppression")



Working Memory ("Maintenance Suppression")



When unwanted memories intrude...



- Intrusions may be graded: retrieval suppression may continue limiting retrieval, but purging what intruded may require a separate process.

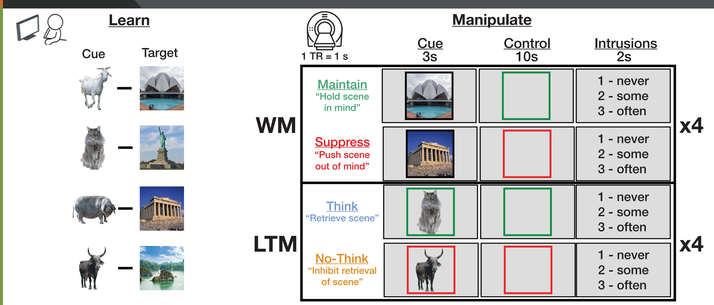
RESEARCH QUESTION

How do maintenance suppression and retrieval suppression interact to regulate intrusive thoughts?

H1: Maintenance suppression and retrieval suppression are distinct control operations that share overlapping mechanisms.

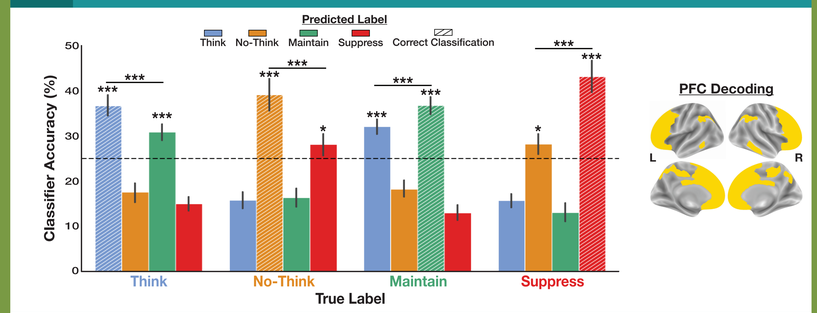
H2: When unwanted memories intrude, maintenance suppression is reactively engaged to purge intrusive content.

2 EXPERIMENTAL APPROACH (N = 45)

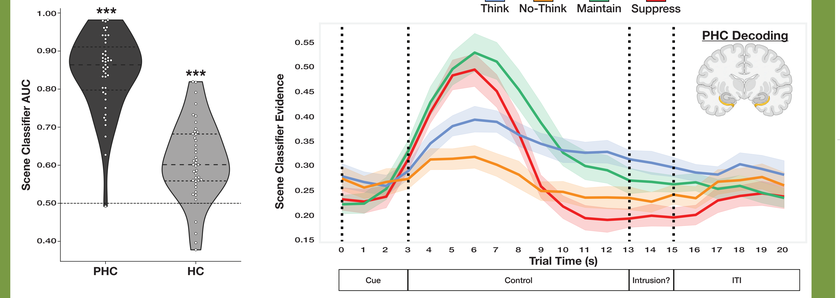


3 DECODING MEMORY CONTROL

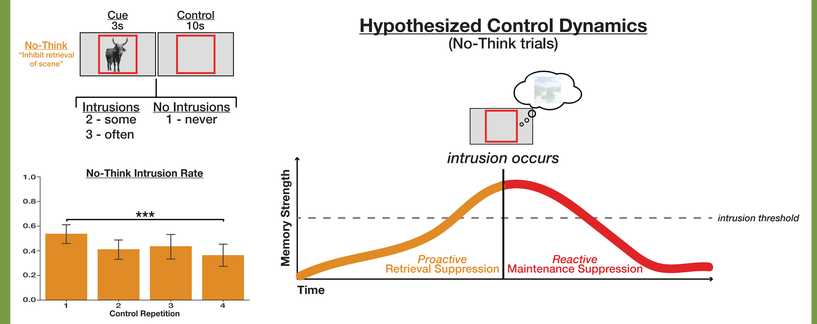
Distinct control operations share common mechanisms



Control shapes content representations

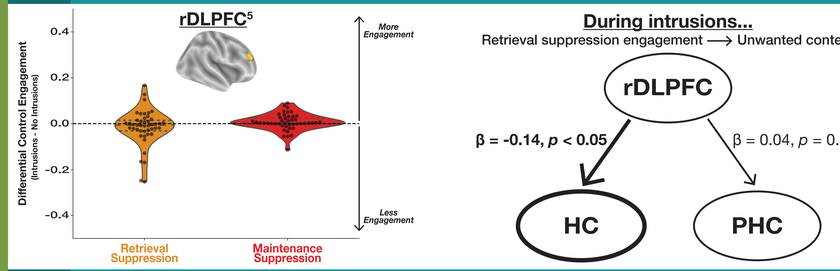


4 RECOVERING FROM INTRUSIONS

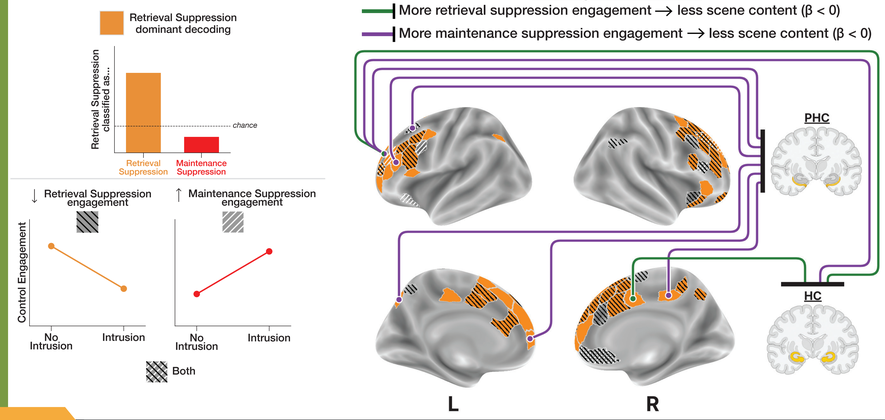


5 CONTROL CHANGES AFTER INTRUSIONS

rDLPFC retrieval suppression only targets HC content



Maintenance suppression backs up retrieval suppression to purge intrusions



6 DISCUSSION

Intrusions flexibly recruit distinct control processes, rather than parallel inhibition by a unitary mechanism.

Maintenance suppression is reactively engaged once a memory intrudes, even in frontal regions dominated by retrieval suppression.

Retrieval suppression keeps preventing further retrieval (HC), while maintenance suppression purges content that already intruded (PHC).

References:

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3. Benoit, R.G., Hulbert, J.C., Huddleston, E., & Anderson, M.C. (2015). *Journal of Cognitive Neuroscience*
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